

Air Force 18 Report:

A Cost-Benefit Analysis of the Maintenance
Situation of F-35s Diverted from Luke Air Force
Base to Gila Bend Auxiliary Airfield

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In conjunction with Luke Air Force Base 56th EMS South AGE Flight

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Executive Summary

F-35 fighter jets from Luke Air Force Base (LAFB) need to divert to nearby Gila Bend Auxiliary Airfield (GBAFAF) 1.5 times per month (on average). While some diversions are simply due to low fuel or poor weather, those due to equipment failure require specialized aerospace ground equipment (AGE) to repair them. The current global AGE pool strategy prohibits the permanent stationing of AGE at GBAFAF, and as a result, it must be transported back and forth between the bases. This results in a large waste of time and money.

The F-35's cousin, the F-16, exemplifies a much better system where the Air Force was able to purchase legacy AGE that is kept ready at both bases. The Air Force 18 Team recommends following a similar strategy by permanently stationing specific AGE at GBAFAF.

Over the year-long period from June 2020 – May 2021, 19 divers caused 20 aircraft to land at GBAFAF. These divers cost LAFB \$27,743.80. Of that, transporting two pieces of AGE, power carts and air carts, was responsible for \$21,692.68 (78%).

The Air Force 18 Team recommends permanently stationing a 270 VDC Power Converter and an Electric Air Conditioner at GBAFAF. Stationing electric variants of both will only cost LAFB \$1,507 annually, resulting in a net savings of \$20,185.82 (73%) per year.

Losing a power cart and an air cart will necessitate the remaining AGE pool increase their workload to replace the lost workhours. However, by having both the electric and diesel variants of the power and air AGE increase their workload, all power AGE will only have to increase their workload by 2.4%, and all air AGE by only 0.49%.

The Air Force 18 Team's proposed solution dramatically reduces transportation costs while nominally increasing maintenance costs and AGE workload, resulting in a far more efficient use of resources.

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I Introduction

Luke Air Force Base (LAFB) in Phoenix, Arizona has the largest concentration of F-35A fighter jets in the world. It serves as the training ground for every F-35A pilot in the Air Force as well as pilots from other nations. As there are so many jets and pilots, semi-frequent emergency landings are needed due to equipment malfunctions, low fuel, and poor weather conditions. These jets are typically diverted to Gila Bend Auxiliary Airfield (GBAFAP) in Gila Bend, Arizona.

The F-35 and supporting aerospace ground equipment (AGE) maintenance strategy relies on a global pool of support equipment that is shared globally for all the variants of the F-35. This methodology requires all assets to be tracked and stationed at each F-35 site, preventing AGE from being strategically positioned off-site for diverted aircraft recovery.

This becomes an issue when F-35s are diverted to GBAFAP due to equipment malfunctions. As AGE cannot be stationed off-site, AGE must be transported from LAFB to GBAFAP, used to repair the plane, and then transported back in a couple of days. This regular (averaged 1.5 times per month) transport of AGE causes significant wait times due to shipment preparation, scheduling logistics, and manpower acquisition. This has developed into a large cost for LAFB and a colossal waste of time.

An alternative arrangement is used for the maintenance of the F-16. The Air Force was able to pre-station legacy AGE which they can keep at convenient locations. Some of this AGE is stationed at GBAFAP. Obviously, this entirely removes associated costs of AGE transport and man hours.

This report uses a cost-benefit analysis to determine possible solutions and recommends the most effective solution to be put into service.

II Data Analysis

All data analyzed in this report comes courtesy of 56th EMS South AGE Flight. To provide recommendations, the Air Force 18 Team performed a cost-benefit analysis using divert data collected over 12 months from June 2020 until May 2021 and AGE usage data collected over 13 months from May 2020 through May 2021.

2.1 Diverts

From June 2020 until May 2021, 26 diverts caused 44 aircraft to land at GBFAF. Of those, 19 diverts totaling 20 aircraft required AGE to prepare the aircraft for flight before returning to LAFB. The team broke these 19 diverts into ten separate categories. These categories were defined by what caused the aircraft to require maintenance. *Figure 1* details the breakdown of the 19 diverts. Note each divert corresponds with one aircraft except for in the fuel issue divert category, in which case one fuel issue divert included two aircraft.

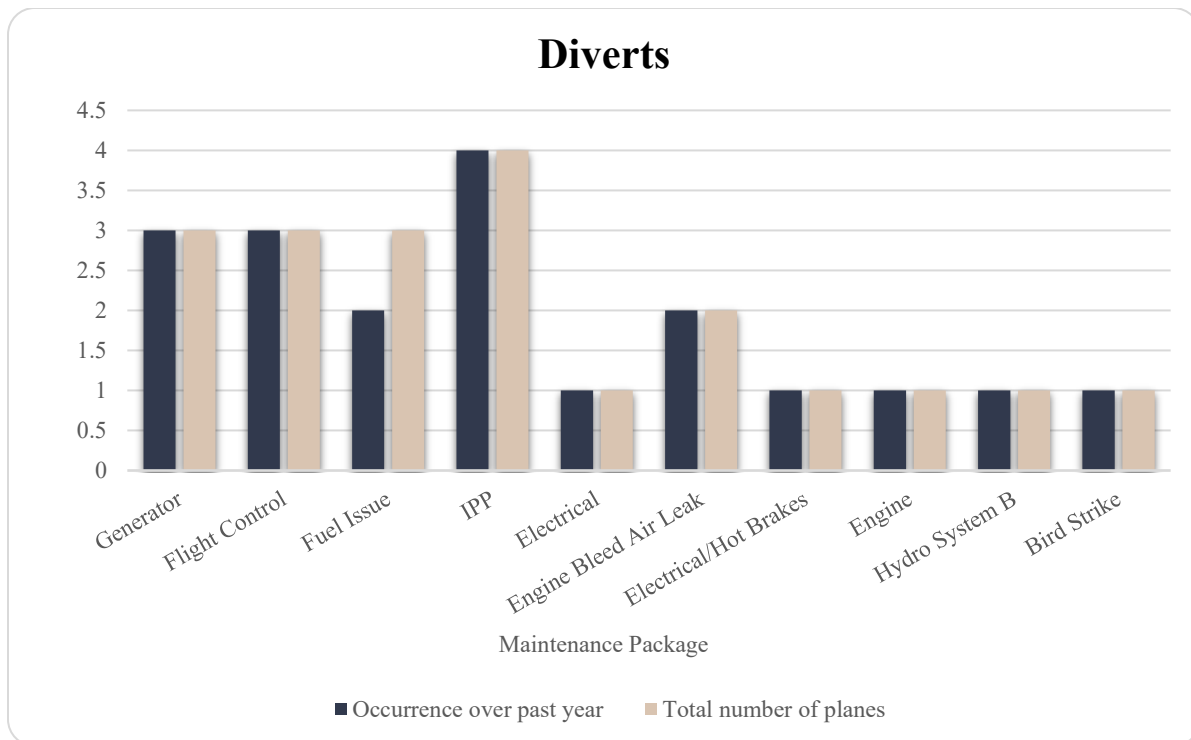


Figure 1. Breakdown of diverts into categories

2.2 Maintenance Packages

Each divert category requires a unique maintenance package to prepare the aircraft for its return flight. These maintenance packages range between two to four pieces of AGE. All maintenance packages require at least power and air AGE to operate the aircraft without starting the jet engine. *Table 1* lists each maintenance package. Each truck transporting AGE can generally

carry two pieces of AGE. The AGE comprising each maintenance package was further divided by the number of trucks needed.

Table 1. Maintenance packages for each divert category

Maintenance Package	AGE Needed					
	Truck 1		Truck 2		No Truck Needed	
Generator	Power	Air				
Flight Control	Power	Air	Hydro Test Stand ¹			
Fuel Issue	Power	Air				
IPP	Power	Air	Component Jack ²	PAO Cart		
Electrical	Power	Air				
Engine Bleed Air Leak	Power	Air				
Electrical/Hot Brakes	Power	Air			Axle Jack	
Engine	Power	Air			Engine Removal Trailer	Engine Transport Trailer
Hydro System B	Power	Air	Hydro Test Stand	Tripod Jack Set ³		
Bird Strike	Power	Air			Engine Removal Trailer	Engine Transport Trailer

As each truck can only carry two AGE carts, some maintenance packages require a second truck to transport all the AGE necessary. Additionally, three pieces of AGE were categorized as not needed to be transported to GBAFAF. The axle jack can fit in the back of a pickup truck, enabling the maintenance crew to carry it with them when driving to GBAFAF, forgoing the need to transport it via flatbed. The engine removal trailer and engine transport trailer are operated by a separate division and are therefore considered in-house, meaning their cost of transportation was outside the scope of this report.

¹ Required 10% of the time.

² At least one, sometimes two, but the second can fit on the same truck.

³ Required only for landing gear maintenance.

2.3 Cost Analysis

2.3.1 Per-Truck Transportation Cost

This report focuses on reducing the cost of transporting AGE to and from GBAFAF for divers. As the personnel cost for maintaining the aircraft will be the same whether the maintenance occurs at LAFB or GBAFAF, the Air Force 18 Team calculated the cost of each divert based off the transportation cost for the AGE.

The transportation cost per truck was calculated using a combination of expenses based on both mileage and total hours. *Table 2* displays the individual expenses and whether they are mileage-based or hourly based. Adding each individual expense gives the total mileage and hourly expenses: \$2.70 and \$34.34, respectively.

Table 2. Cost breakdown per mile and per hour

Item	Cost per mile	Cost per hour
Chains/Binders/Straps	\$0.41	\$0
PPE (steel-toe boots/gloves)	\$1.75	\$0
Fuel (Diesel)	\$0.53	\$0
Truck maintenance	\$0.01	\$0
Personnel (2)	\$0	\$34.34
Totals	\$2.70	\$34.34

The round trip from LAFB to GBAFAF and then back covers 122.4 miles and takes 7 hours—including the time to load and offload the AGE. This gives a cost breakdown of $\$2.70 * 122.4 = \330.48 for total cost based off mileage and $\$34.34 * 7 = \240.38 for total cost based off time, giving a total cost of \$570.86 for one truck to drive from LAFB to GBAFAF and back. *Figure 2* provides a visual representation of this cost breakdown.

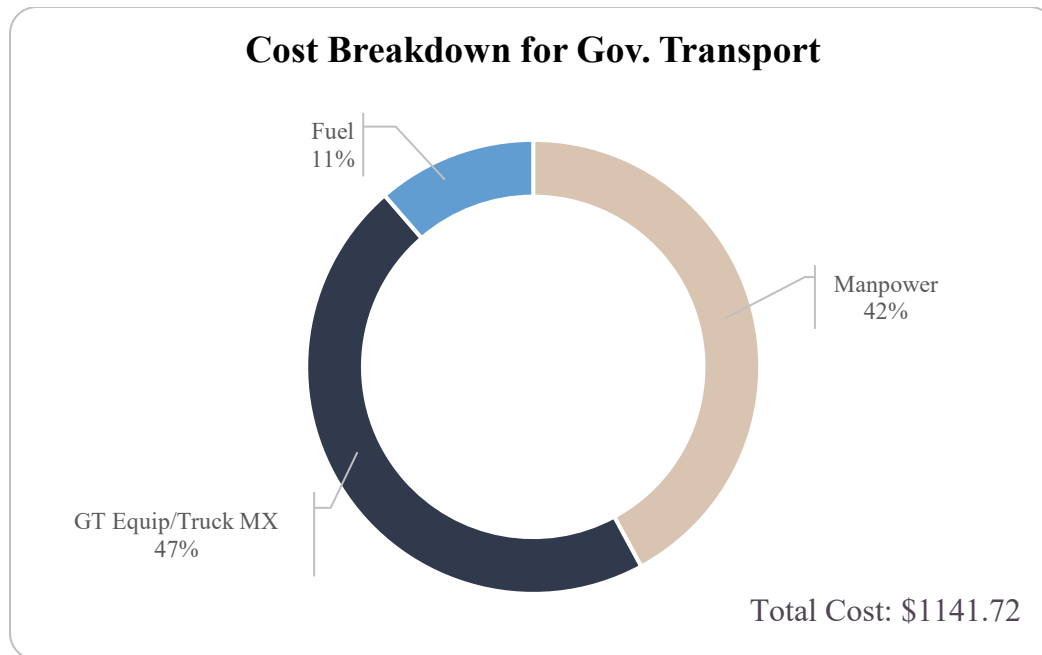


Figure 2. Cost breakdown for government transport of AGE

To prepare an aircraft for flight at GBAFAF, AGE must be transported from LAFB to GBAFAF. Then, after the aircraft is flight-ready, the AGE must be transported back from GBAFAF to LAFB. As these occur on separate days, the trucks transporting the AGE must travel from LAFB to GBAFAF and then back to LAFB for each divert *twice*. They travel once to drop the AGE off and then again to pick it up, meaning one truckload of AGE costs $\$570.86 * 2 = \1141.72 to transport.

2.3.2 Maintenance Package Costs

As each truck can transport two pieces of AGE, the cost of each maintenance package can be calculated by dividing the number of pieces of AGE requiring transport on a truck by two, rounding up to the nearest whole number, and then multiplying this number by the cost of transporting one truckload of AGE (\$1141.72).

The only exception to this formula is the flight control maintenance package. The flight control issue requires a hydro test stand ten percent of the time—and therefore requires a second transportation truck ten percent of the time since the hydro test stand is the third piece of AGE for this maintenance package. The team calculated the cost by averaging the number of trucks to 1.1, giving a maintenance package cost of \$1255.89.

To calculate the total cost per year of each maintenance package, the team multiplied the per-divert maintenance-package cost by the occurrence of each divert (see *Figure 1* for the total

occurrence for each divert category). *Figure 3* charts the per-divert and yearly cost of each maintenance package.

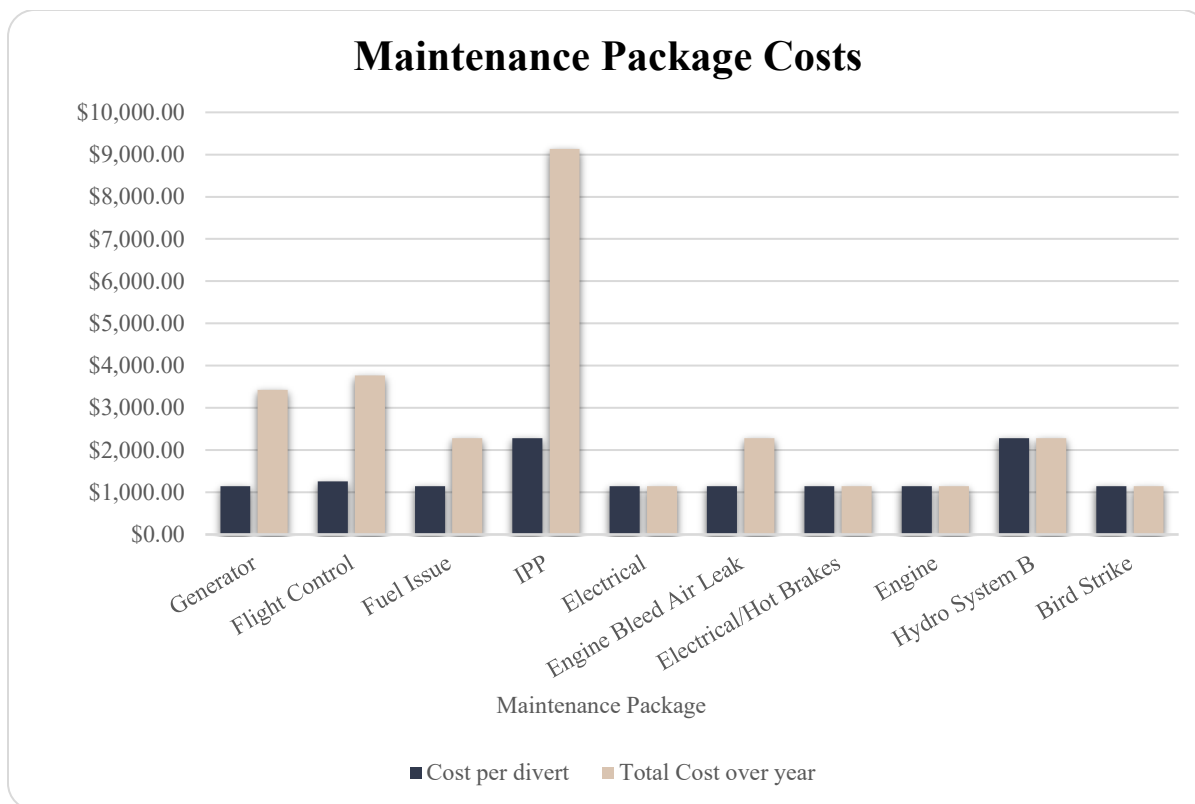


Figure 3. Cost of each divert category and total cost over the 12-month period

In total, transporting AGE to GBAFAF airfield to service diverted aircraft costs LAFB \$27,743.80 per year. This expense can be drastically reduced by strategically stationing AGE at GBAFAF.

III Proposed Solution

As *Figure 4* clearly shows, most of the total expenses consist of transporting the power and air AGE. In fact, this makes up 78% (\$21,692.68) of the total cost.

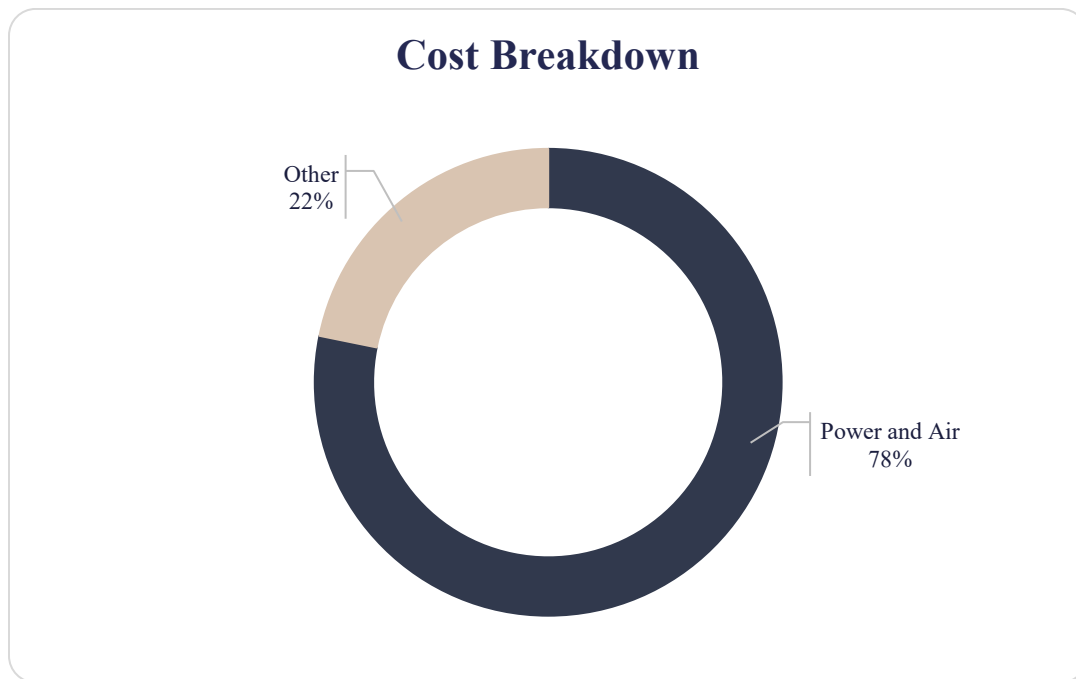


Figure 4. Cost breakdown chart

The Air Force 18 Team recommends permanently stationing a 270 VDC Power Converter and an Electric Air Conditioner at GBFAF.

LAFB utilizes both diesel and electric variants of their AGE. Diesel AGE runs off a diesel tank attached to the equipment, and electric AGE must be plugged into facility power or a generator. The team recommends electric variants because GBFAF has two facility power outlets built into the maintenance hangar which the AGE can use.

Permanently stationing this AGE at GBFAF will dramatically reduce the expenses related to transporting AGE to GBFAF to prepare diverted aircraft for their flight to LAFB, saving LAFB a projected \$21,692.68 per year with only minimal drawbacks which are discussed in depth below as well as potential solutions to the costs of implementation.

3.1 Cost Reduction

Stationing power and air AGE at GBFAF will reduce the cost for each maintenance package.

Figure 5 displays these cost reductions for the total expenses associated with each maintenance package.

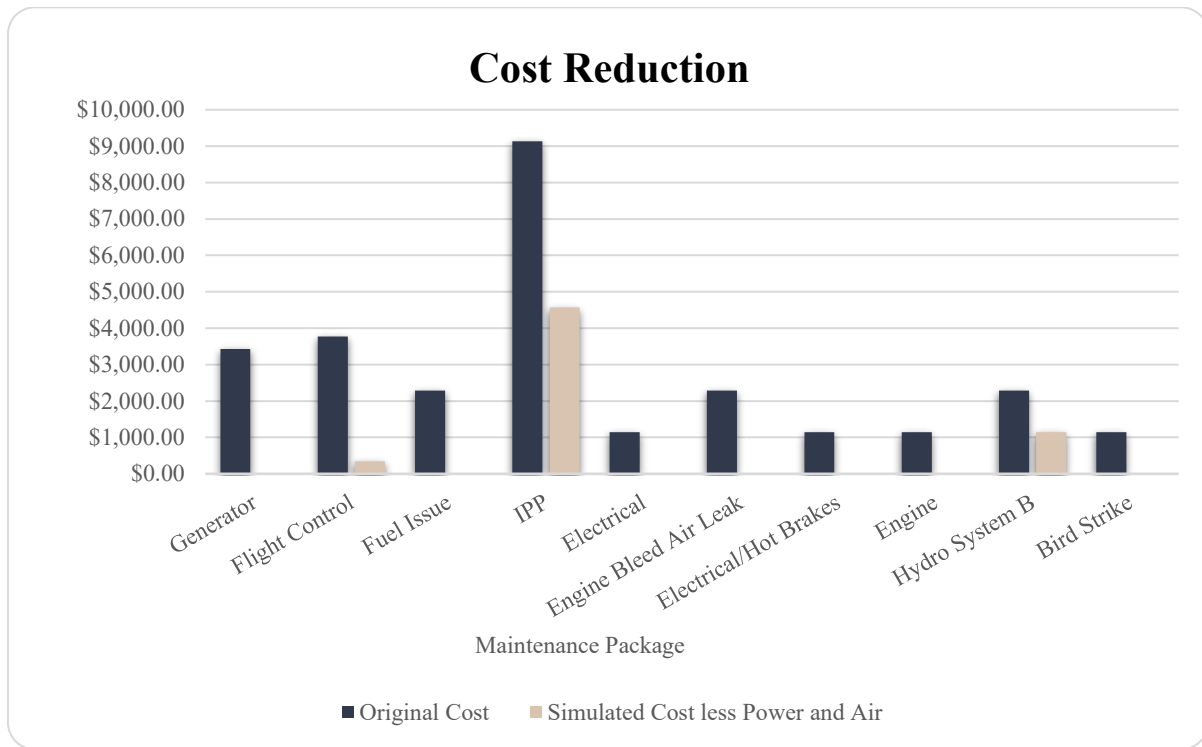


Figure 5. Cost reduction of proposed solution for each maintenance package

Of the ten maintenance packages, two will see a 50% cost reduction, one a 90% reduction, and seven a 100% reduction. The most expensive maintenance package, IPP, will only see a 50% cost reduction, but the overwhelming need for power and air AGE enables a 78% overall cost reduction.

3.2 Cost of Implementation

Stationing power and air AGE at GBFAF will incur its own expenses. The overall savings will immensely outweigh these additional expenses, but they must be considered for a more holistic understanding of the proposed solution.

The AGE stationed at GBFAF will require bi-weekly maintenance and inspection. As they would normally undergo these while stationed at LAFB, the costs of manpower and maintenance will be the same; however, a crew of two engineers will need to travel from LAFB to GBFAF and back in a government vehicle to perform the maintenance and inspection. Given a fuel-tank size of twelve gallons and a gas price of \$3, the cost of this bi-weekly maintenance will be \$36 per trip. Since the bi-weekly maintenance will occur 26 times throughout the year, the yearly cost will be $\$36 * 12 = \936 .

The AGE will be switched out once a year with fresh AGE from LAFB. This will enable the AGE technicians to perform the preventative maintenance on the AGE. Switching the stationed AGE will necessitate transportation costs. As the stationed AGE can be transported back to LAFB on the same day and by the same truck by which the new AGE was transported, the cost of switching out the AGE is the same as a one-way delivery of equipment (\$570.86, see section 2.3.1 Per-Truck Transportation Cost).

The bi-weekly and yearly expenses combine to \$1507.86 per year, far less than the \$21,692.68 saved by the proposed solution. These expenses cause the net savings of implementing the Air Force 18 Team's proposed solution to equal \$20,185.82 per year which constitutes 73% of the current annual expenses for transporting AGE to and from GBAFAF.

3.3 Effects on AGE Needs at LAFB and Proposed Solutions

Any F-35 AGE stationed at GBAFAF will have to be pulled from LAFB's collective pool. This means LAFB will have one less electric power cart and one less electric air cart. Currently, a total of 26 power carts (ten electric and sixteen diesel) and 29 air carts (twelve electric and seventeen diesel) are stationed at LAFB. The proposed solution will reduce the number of available electric power AGE to nine and electric air AGE to eleven.

Table 3 displays the AGE usage data for these four variants. The diesel variants see a greater workload in comparison with their electric counterparts, and the electric power cart receives the least use of the four.

Table 3. Power and air AGE usage data

Type of AGE	Number in service	Total Hours	Hours per Cart per Active Month
Cart, Con Air Diesel	17	4239	21.30
Cart, Con Air Electric	12	1618	10.51
Cart, 270 VDC, Diesel	16	5013	25.58
Converter, Power, 270 VDC	10	254.4	1.988

Stationing power and air AGE at GBAFAF will result in a loss of 10.5 workhours for air AGE and 1.99 workhours for power AGE per month. This loss of workhours must be recuperated somehow, and the Air Force 18 Team proposes three possibilities, described in depth below.

3.3.1 Replace with New AGE

Replacing the stationed AGE with new AGE presents itself as the simplest solution. A new electric power cart costs \$100,589, and a new electric air cart costs \$117,599, summing to a total

replacement cost of \$218,188. Given a net savings of \$20,185.82 per year, it would take LAFB 10.81 years of implementing the proposed solution to offset the replacement cost. Given the length of payoff period, the low percentage of workload lost, and the inability of the Department of Defense to outright buy F-35 AGE, the Air Force 18 Team does not recommend this solution for recuperating the lost workhours.

3.3.2 Have On-Base AGE Increase Workload

The remaining AGE can increase their workload to make up for the loss of workhours. This can be implemented via a variety of different strategies. The team chose to explore two logical strategies based on the problem: have only electric variants increase their workload or have both diesel and electric variants increase their workload by the same percentage.

3.3.2.1 *Electric Only*

Having the eleven remaining electric variants of the air AGE and the nine remaining of the power AGE increase their workload to compensate for the missing AGE offers the most seamless transition because the type of AGE performing the job would not change. However, this strategy would result in a sizable increase in the remaining carts' workload. *Table 4* details this workload increase. This strategy would result in the electric variant of the air AGE increasing its monthly workload by 9.1% and the electric variant of the power AGE by 11%.

Table 4. Simulated numbers for electric-only strategy

Type of AGE	Simulated Hours per Month	Increase (Hours)	Percent Inc.
Cart, Con Air Diesel	21.30	0	0
Cart, Con Air Electric	11.46	0.9552	9.091%
Cart, 270 VDC, Diesel	25.58	0	0
Converter, Power, 270 VDC	2.208	0.2208	11.11%

3.3.2.2 *Both Diesel and Electric with Same Increase by Percentage*

A more complicated but better-balanced approach—having the diesel and electric variants increase their workload by the same percentage—limits the change in workload for all the AGE. *Table 5* explores this possibility. As the diesel variants regularly experience more monthly use, this strategy enables the workload percentage increase to stay quite low, with both the diesel and electric variants of the air and power AGE seeing meager increases in their monthly workload. The air AGE would increase their workload by 2.4% and the power by 0.49%.

Table 5. Simulated numbers for even-split strategy

Type of AGE	Simulated Hours per Month	Increase (Hours)	Percent Inc.
Cart, Con Air Diesel	21.80	0.5020	2.357%
Cart, Con Air Electric	10.75	0.2476	2.357%
Cart, 270 VDC, Diesel	25.70	0.1241	0.4853%
Converter, Power, 270 VDC	1.997	0.009645	0.4853%

IV Conclusion

Implementing the proposed solution of permanently stationing electric variants of both power and air AGE at GBAFAP will dramatically reduce LAFB's expenses for transporting AGE to service diverted F-35 aircraft. LAFB will save \$21,692.68 (78%) per year while only spending an additional \$1507.86, resulting in a net savings of \$20,185.82 (73%) annually.

Stationing these two pieces of AGE will cause a reduction in the AGE-pool size for power and air AGE. However, by having both the electric and diesel variants of the power and air AGE increase their workload to replace the lost workhours, all power AGE will only have to increase their workload by 2.4% and all air AGE only 0.49%.

The proposed solution should be implemented as soon as possible to capitalize on the possible cost reductions and reduce unnecessary transportation of AGE.

V About the Authors

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